

Methods

Overloading

Produced Dr. Siobhán Drohan
by: Mr. Colm Dunphy
 Ms Mairead Meagher
 Ms Siobhan Roche

Topics list

1. RECAP: Method Terminology:
2. Method examples – return values
 - Celcius / Farenheit **Converter**.
 - **Cubed**
 - **Product**
3. **Overloading** methods.
4. **Recursion**.

Overloaded methods

- Multiple methods can have the **same name**, once they have a **different parameter list**.
- We could write the following:

– void addNum()	
– void addNum (int num1, int num2)	
– void addNum (int num1, int num2, int num3)	
– void addNum (int num1, int num2, int num3, int num4)	

Same Name

Different Parameter List

Overloaded methods

Method signature	Parameter List
<code>void addNum()</code>	no parameter
<code>void addNum(int num1, int num2)</code>	int, int
<code>void addNum(int num1, int num2, int num3)</code>	int, int, int
<code>void addNum(int num1, int num2, int num3, int num4)</code>	int, int, int, int

Same method name, but different parameter list

Overloaded methods

- A program can have two or more methods with the same name, only if their parameter list is different.
- When Java is checking that a parameter list is different, it is not checking the name of the variables, it is **checking the data type** of the variables
e.g.
 - this is permitted as the **data type** is different:
 - void addNum (int num1)
 - void addNum (double num1)

Data types must be different
for a method with the same name to be overloaded

Overloaded methods

```
public class Driver {  
  
    public static void main(String[] args) {  
        ClassExamples ce = new ClassExamples();  
        double cel = ce.fahrenheitToCelsius(451);  
        double c = 20.5;  
        double d = 3.5;  
        ce.addNum(c, d);  
    }  
}
```



Q: Which addNum method is called and why?

```
public class ClassExamples {  
  
    void addNum(double num1, double num2){  
        double result = num1 + num2;  
        System.out.println("This is double addNum " +  
result);  
    }  
  
    void addNum(int num1, int num2){  
        int result = num1 + num2;  
        System.out.println("This is int addNum " + result);  
    }  
  
}
```

```
This is double addNum 24.0
```

Overloaded methods

- When you call a method, Java **matches** the **number and type of the arguments** you passed to the method, with all the declared methods.
- When a match is found, Java invokes that method
e.g.

`addNum (0, 4)` calls `void addNum (int num1, int num2)`

`addNum (0.0, 4.0)` calls `void addNum (double num1, double num2)`

Key features of Method Overloading

- Multiple methods can share the same name in a class when their parameter lists are different.
- Overloading is a way to increase flexibility and improve the readability of code.
- Overloading does not depend on the return type of the method, two methods cannot be overloaded by just changing the return type.
- It does depend on:
 - The number of parameters
 - The types of parameters
 - The order of parameters

Questions?

